

- 2 Scientists have found very little evolutionary change in populations of two Australian songbirds, the zebra finch, *Taeniopygia guttata castanotis*, and the budgerigar, *Melopsittacus undulatus*.

(a) Describe the process of evolution by natural selection.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(b) The number of eggs a bird lays in its nest is called the clutch size.

The variation in clutch size was investigated in the zebra finch over several years.

The data are shown in Fig. 2.1.

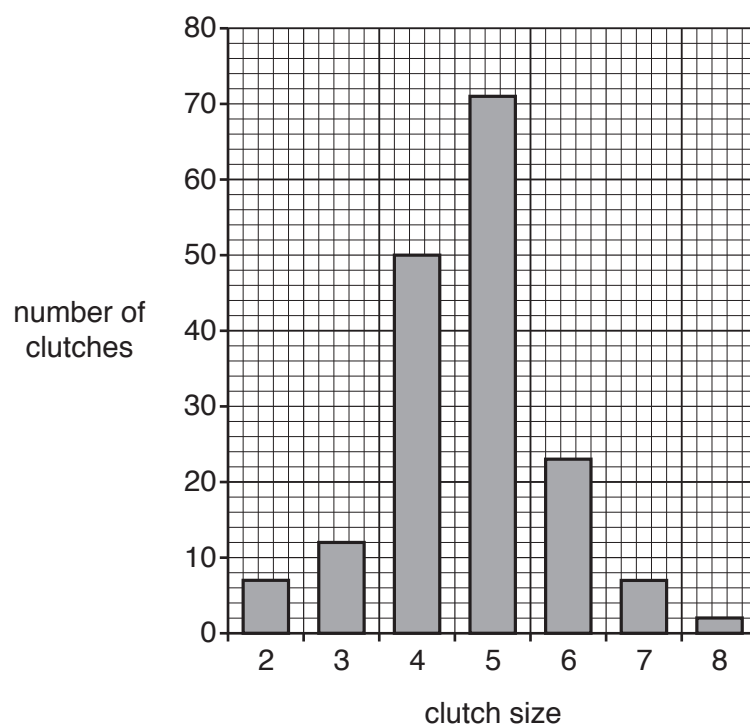


Fig. 2.1

- (i) Describe the pattern shown by the data in Fig. 2.1.

.....

.....

.....

.....[2]

- (ii) The data in this investigation were collected over 60 years ago.

The same investigation, carried out today, would produce the same pattern of results.

Explain how the selection factors acting on zebra finches would maintain the same pattern of results.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(c) The Hardy–Weinberg principle is used to calculate allele, genotype and phenotype frequencies in populations.

- A breeder of birds keeps a population of 86 budgerigars in one enclosed area.
- Two distinct phenotypes are present, blue feathers and green feathers.
- Feather colour is controlled by one gene:
 - **G** is the allele for green feathers
 - **g** is the allele for blue feathers.
- Only 17 of the budgerigars have blue feathers.

(i) The Hardy–Weinberg equations are shown in Fig. 2.2.

$$p + q = 1$$
$$p^2 + 2pq + q^2 = 1$$

Fig. 2.2

Calculate the number of heterozygous individuals in the population.

Show your working.

number =[3]

(ii) The Hardy–Weinberg principle cannot be applied to all populations.

State **two** conditions when the Hardy–Weinberg principle **cannot** be applied.

- 1
-
- 2
-[2]

[Total: 15]